

Name : Sajid Hasan

ID : 2105145

Dept : CSE

Section: C

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For, a , $145 \bmod 3 = 1$

For b , $145 \bmod 5 = 0$

(c)

Definition : $H = \{ (M, w) \mid M \text{ halts on } w \}$

Halting Problem is RE:

- (i) We can construct a TM U_H which takes as input a TM M and a string w
- (ii) U_H simulates the action of M on w
- (iii) If M halts, U_H accepts and halts
- (iv) But if M keeps running, U_H also keeps running

A language L is RE if there is a TM that can accept any string in L , but may run forever for strings not in L .

According to (iii) & (iv), H has the aforementioned properties. Therefore, the Halting problem is Recursively Enumerable (RE).

Halting Problem is not recursive:

(i) We know that L_u is not recursive

(ii) Suppose for contradiction that the halting problem is recursive.

iii) Now, to determine whether a TM M accepts w , we could do the following,

→ Determine whether M halts on w .

→ If it does, execute M on w and see whether M halts in an accepting state. If so, accept (M, w)

→ If M doesn't halt on w , reject (M, w)

iv) Thus, we could test whether M accepts w by using our assumed algorithm for determining halting. We have "reduced L_A to the halting problem".

⑤ So, if the halting problem is recursive, then so is L_u . But we know that L_u is not recursive, so, the assumption that the halting problem is recursive must be false.

(a)

(1)

In the next page:

$$\{a^n b^n c^m d^m \mid n, m \geq 1\} \cup \{a^n b^m c^m d^n \mid n, m \geq 1\}$$

